WORKED SOLUTION

STEP III 2013, Question 13

Probability and Statistics • Cdf

STEP III 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q13

The key idea

The expression $2\mathrm{F}(x)\mathrm{f}(x)$ is the derivative of $(\mathrm{F}(x))^2$, because $\mathrm{F}' = \mathrm{f}$. That makes part (i)(b) a single integration by parts, and with the right choice of g it turns every question about Y into a question about X. The bounds then come from replacing f or F by the bigger quantities M or Mx from part (i)(a), each time in the direction that makes the integral larger.

Solution

Throughout, $\mathrm{F}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$, so $\mathrm{F}'(x) = \mathrm{f}(x)$, with $\mathrm{F}(0) = 0$ and $\mathrm{F}(1) = 1$ since X lies in $[0, 1]$.

Part (i)

(a) For $0 \le x \le 1$, since $0 \le \mathrm{f}(t) \le M$ for every t,

$$0 \le \int_0^x \mathrm{f}(t)\,\mathrm{d}t \le \int_0^x M\,\mathrm{d}t = Mx,$$

which is $0 \le \mathrm{F}(x) \le Mx$.

(b) By the chain rule, $\dfrac{\mathrm{d}}{\mathrm{d}x}(\mathrm{F}(x))^2 = 2\mathrm{F}(x)\mathrm{f}(x)$. Integrating by parts with $u = \mathrm{g}(x)$ and $\mathrm{d}v = 2\mathrm{F}(x)\mathrm{f}(x)\,\mathrm{d}x$, so $v = (\mathrm{F}(x))^2$,

$$\begin{aligned}\int_0^1 2\mathrm{g}(x)\mathrm{F}(x)\mathrm{f}(x)\,\mathrm{d}x &= \Big[\mathrm{g}(x)(\mathrm{F}(x))^2\Big]_0^1 - \int_0^1 \mathrm{g}'(x)(\mathrm{F}(x))^2\,\mathrm{d}x \\ &= \mathrm{g}(1) - \int_0^1 \mathrm{g}'(x)(\mathrm{F}(x))^2\,\mathrm{d}x,\end{aligned}$$

since $\mathrm{F}(1) = 1$ and $\mathrm{F}(0) = 0$. (We need g differentiable, which the question implicitly assumes.)

Part (ii)

(a) The density of Y must integrate to 1. Taking $g(x) = 1$ in (i)(b), so that $g' = 0$,

$$\int_0^1 2F(y)f(y)\,dy = 1,$$

so $k = 2$. (Equally, $\int_0^1 kFf\,dy = \frac{k}{2}[F^2]_0^1 = \frac{k}{2}$.) Note also that $kFf \geq 0$, so this really is a density.

(b) We have

$$E(Y^n) = \int_0^1 2y^n\,F(y)f(y)\,dy.$$

Upper bound. By (i)(a), $F(y) \leq My$, and the other factors $2y^n f(y)$ are non-negative, so

$$E(Y^n) \leq \int_0^1 2y^n \cdot My \cdot f(y)\,dy = 2M\int_0^1 y^{n+1}f(y)\,dy = 2M\mu_{n+1}.$$

Lower bound. Now use (i)(b) with $g(y) = y^n$, so $g(1) = 1$ and $g'(y) = ny^{n-1}$:

$$E(Y^n) = 1 - \int_0^1 ny^{n-1}(F(y))^2\,dy.$$

(When $n = 0$ the integral is simply 0.) To bound $E(Y^n)$ from below we need to bound this integral from *above*.

? **Ask yourself** Which of the two factors of $(F(y))^2$ should we replace by My?

Just one of them. Since $0 \leq F(y) \leq My$, we have $(F(y))^2 \leq My\,F(y)$, and $ny^{n-1} \geq 0$, so

$$\int_0^1 ny^{n-1}(F(y))^2\,dy \leq nM\int_0^1 y^n F(y)\,dy.$$

Replacing both factors would leave $\int y^{n+1}\,dy$, which says nothing about X; keeping one F lets us integrate by parts once more and produce a moment of X. Integrating by parts with $u = F(y)$ and $dv = y^n\,dy$,

$$\int_0^1 y^n F(y)\,dy = \Big[\frac{y^{n+1}}{n+1}F(y)\Big]_0^1 - \int_0^1 \frac{y^{n+1}}{n+1}f(y)\,dy = \frac{1-\mu_{n+1}}{n+1}.$$

Putting this together,

$$E(Y^n) \geq 1 - \frac{nM}{n+1}(1-\mu_{n+1}) = 1 + \frac{nM}{n+1}\mu_{n+1} - \frac{nM}{n+1},$$

which, with the upper bound, is the required pair of inequalities.

(c) Combining the two ends of (b), for $n \geq 0$,

$$1 + \frac{nM}{n+1}\mu_{n+1} - \frac{nM}{n+1} \leq 2M\mu_{n+1},$$

and collecting the μ_{n+1} terms on the right,

$$1 - \frac{nM}{n+1} \leq \left(2M - \frac{nM}{n+1}\right)\mu_{n+1} = \frac{(n+2)M}{n+1}\,\mu_{n+1}.$$

The coefficient $\frac{(n+2)M}{n+1}$ is positive ($M > 0$, since f integrates to 1), so we may divide by it without reversing the inequality:

$$\mu_{n+1} \geq \frac{n+1}{(n+2)M} - \frac{n}{n+2}.$$

This holds for every $n \geq 0$. Writing $m = n+1$, so that $m \geq 1$, it becomes

$$\mu_m \geq \frac{m}{(m+1)M} - \frac{m-1}{m+1} \qquad (m \geq 1),$$

which is the required result (with m renamed n).

Checks

- Uniform X: $\mathrm{f} = 1$, $M = 1$, $\mu_n = \frac{1}{n+1}$. The bound in (c) is $\frac{n}{n+1} - \frac{n-1}{n+1} = \frac{1}{n+1}$, so it holds with equality. Here Y has density $2y$ and $\mathrm{E}(Y^n) = \frac{2}{n+2}$, and both bounds in (b) also equal $\frac{2}{n+2}$. This makes sense: every inequality we used, $\mathrm{F}(y) \leq My$, is an equality for the uniform distribution.
- $\mathrm{f}(x) = 3x^2$, $M = 3$, $n = 2$: $\mu_2 = \frac{3}{5}$ and the bound is $\frac{2}{9} - \frac{1}{3} < 0$, comfortably true. For this density, $\mathrm{E}(Y^2) = \frac{3}{4}$ and $\mu_3 = \frac{1}{2}$, so the bounds in (b) are $1 + 2 \cdot \frac{1}{2} - 2 = 0$ and $2 \cdot 3 \cdot \frac{1}{2} = 3$, and $\frac{3}{4}$ lies between them.
- The identity in (i)(b) and all the inequalities were also confirmed by numerical integration for several densities and several choices of g.

Common slips

- Bounding $\left(\mathrm{F}\right)^2$ by M^2y^2: that is true but leads nowhere, since it loses all information about X.
- Forgetting that $\mathrm{E}(Y^n)$ appears with a minus sign in front of the integral, so a lower bound for $\mathrm{E}(Y^n)$ needs an *upper* bound for the integral.
- Stopping at the bound for μ_{n+1} without shifting the index to get the stated form for $n \geq 1$.

Beyond STEP

The density $2\mathrm{F}(y)\mathrm{f}(y)$ is no accident: it is the density of $\max(X_1, X_2)$ for two independent copies of X, since $\mathrm{P}(\max \leq y) = \big(\mathrm{F}(y)\big)^2$. More generally, the rth smallest of n independent observations has a density built from powers of F and $1 - \mathrm{F}$; these *order statistics* are central to non-parametric statistics. Blitzstein and Hwang, *Introduction to Probability*, has a clear treatment of order statistics.

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